Prayas JEE AIR (2025)

WEEKLY TEST 23

DURATION : 180 Minutes | **DATE : 22/12/2024** | **M. MARKS : 300**

ANSWER KEY

PHYSICS

1.	(3)
2.	(2)
3.	(2)
4.	(4)
5.	(2)
6.	(2)
7.	(4)
8.	(1)
9.	(4)
10.	(2)
11.	(1)
12.	(4)
13.	(2)
14.	(4)
15.	(2)
16.	(2)
17.	(3)
18.	(1)
19.	(2)
20.	(2)
21.	(20)
22.	(3)
23.	(20)
24.	(600)
25.	(220)

CHEMISTRY

26.	(2)
27.	(4)
28.	(2)
29.	(3)
30.	(1)
31.	(2)
32.	(2)
33.	(3)
34.	(2)
35.	(3)
36.	(2)
37.	(1)
38.	(3)
39.	(3)
40.	(1)
41.	(2)
42.	(3)
43.	(3)
44.	(1)
45.	(1)
46.	(56)
47.	(7)
48.	(5)
49.	(14)
50.	(526)

MATHEMATICS

51.	(1)
52.	(4)
53.	(3)
54.	(2)
55.	(1)
56.	(1)
57.	(3)
58.	(3)
59.	(3)
60.	(4)
61.	(1)
62.	(1)
63.	(1)
64.	(1)
65.	(3)
66.	(4)
67.	(2)
68.	(2)
69.	(1)
70.	(4)
71.	(15)
72.	(31)
73.	(4)
74.	(1)
75.	(3)

SECTION-I (PHYSICS)

1. **(3)**

Two surfaces are formed in the capillary

Since force due to surface tension becomes twice

$\therefore\ h' = 2h$

2. **(2)**

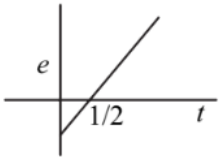
$I = I_0 t - I_0 t^2$

$\phi = BA$

$\phi = \mu_0 nIA$

$V_R = \dfrac{d\phi}{dt} = -\mu_0 nAI_0(1-2t)$

$V_R = 0$ at $t = \dfrac{1}{2}$ and $I_R = \dfrac{V_R}{\text{Resistance of loop}}$



3. **(2)**

$\Delta KE = \Delta PE$ and $\overrightarrow{p_i} = \overrightarrow{p_f}$

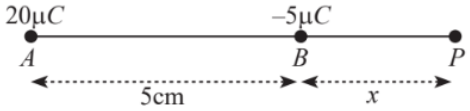
$\dfrac{J^2}{2m} = \dfrac{1}{2}kx^2 + \dfrac{J^2}{2(5\,m)}$

$\Rightarrow\quad x = \dfrac{2\,J}{\sqrt{5\,mk}} = \dfrac{2\times 10}{\sqrt{5\times 1\times 1000}}$

$= \dfrac{2}{\sqrt{50}} = \dfrac{\sqrt{2}}{5}\,\text{m}$

4. **(4)**

From observation, we can say that net force on the charge particle at point *P* is zero.



So, $\dfrac{k(5\mu C)q}{x^2} = \dfrac{k(20\mu C)q}{(5+x)^2}$

$\Rightarrow \dfrac{1}{x^2} = \dfrac{4}{(5+x)^2}$

$\Rightarrow \left(\dfrac{5+x}{x}\right)^2 = 2^2$

$\Rightarrow \dfrac{5+x}{x} = 2$

$\Rightarrow x = 5\,\text{cm}$

5. **(2)**

$\dfrac{1}{2}Li^2 = \dfrac{1}{n}\dfrac{1}{2}Li_0^2$

$\Rightarrow\quad i = \dfrac{i_0}{\sqrt{n}}$

During growth of current,

$i = i_0\left(1 - e^{-t/\tau}\right)\left(\tau = \dfrac{L}{R}\right)$

$e^{t/\tau} = \dfrac{\sqrt{n}}{\sqrt{n}-1}$

$t = \dfrac{L}{R}\ln\left(\dfrac{\sqrt{n}}{\sqrt{n}-1}\right)$

6. **(2)**

We have, $W = \alpha^2\beta e^{-\frac{\beta^2}{kT}}$

$\therefore \left[\beta x^2\right] = [kT] = \left[ML^2T^{-2}\right]\left[\therefore E = \dfrac{3}{2}kT\right]$

$\Rightarrow [\beta] = \left[MT^{-2}\right]$

And, $[W] = \left[\alpha^2\beta\right] \Rightarrow \left[ML^2T^2\right] = \left[\alpha^2\right]\left[MT^{-2}\right]$

$\Rightarrow [\alpha] = [L]$

7. **(4)**

Electric field $t = 0\ \ \&(x, y, z) = \left(0, 0, \dfrac{\pi}{k}\right)$

$\vec{E} = -\left(\dfrac{\hat{i}+\hat{j}}{\sqrt{2}}\right)E_0$ and $\vec{F} = \vec{E}q$

Due to electric field $\vec{F} = -\left(\dfrac{\hat{i}+\hat{j}}{\sqrt{2}}\right)q$

Hence net force is antiparallel to $\left(\dfrac{\hat{i}+\hat{j}}{\sqrt{2}}\right)$.

8. **(1)**

Moment of inertia of solid sphere about any tangent,

$I = I_{cm} + mr^2 \dfrac{2}{5}mr^2 + mr^2$

$= \dfrac{7}{5}mr^2$

Moment of inertia of hollow sphere about any tangent,

$I = I_{\text{cm}} + mr^2$

$= \dfrac{2}{3}mr^2 + mr^2 = \dfrac{5}{3}mr^2$

M.O.I of circular ring about its diameter,

$I = \dfrac{MR^2}{2}$

M.O.I of circular disc about any diameter,

$I = \dfrac{MR^2}{4}$

9. **(4)**

Both voltmeters are parallel.

Both the voltmeters *A* and *B* are effectively in parallel and hence give the same reading every time.

10. **(2)**

We have,

$x = 3t$ $\quad y = 5t^3$

$V_x = \frac{dx}{dt} = 3$ $\quad V_y = \frac{dy}{dt} = 15t^2$

$a_x = \frac{dV_x}{dt} = 0$ $\quad a_y = \frac{dV_y}{dt} = 30t$

$\frac{1}{a} = a_x\hat{x} + a_y\hat{y} + a_z\hat{z}$

Since, *z* coordinate is constant. Therefore, $a_z = 0$

$\vec{a} = 30t\hat{j}$

at t = 1sec $\quad \vec{a} = 30\hat{j}$

11. **(1)**

We know that a dipole is a pair of equal and opposite charges separated by a small distance.

$\therefore$The net charge on the electric dipole, $q_{net} = +q - q = 0$

Therefore, according to Gauss's law,

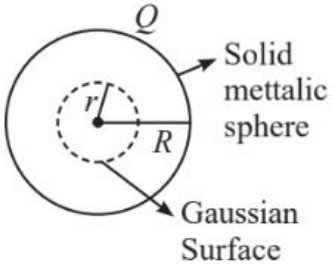
Electric flux, $\oint \vec{E} \cdot \vec{ds} = \frac{q_{net}}{\varepsilon_0} = \frac{0}{\varepsilon_0} = 0$

Where, $\vec{E}$ is the electric field, $\vec{ds}$ is the surface area a small section, and ε_0 is permittivity in free space.

And for a small section $\vec{ds}$ only, $\vec{E} \cdot \vec{ds} \neq 0$

$\therefore$ The flux of the electric field throughout the hollow sphere is zero and the electric field is non-zero. Hence, statement-I is true:

Electric field in region ($r < R$)



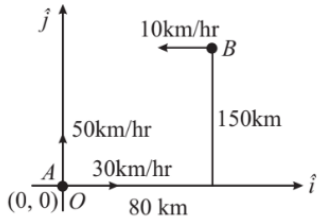
As change encloses within the gaussian surface is equal to zero such as.

$\phi = \oint \vec{E} \cdot \vec{ds}$

Hence, Statement-II is false:

12. **(4)**

If we take the position of ship '*A*' as origin then positions and velocities of both ships can be given as:



$\vec{v}_A = (30\hat{i} + 50\hat{j})$km / hr

$\vec{v}_B = -10\hat{i}$ km / hr

$\vec{r}_A = 0\hat{i} + 0\hat{j}$

$\vec{r}_B = (80\hat{i} + 150\hat{j})$km

Time after which distance between them will be minimum

$t = -\frac{\vec{r}_{BA} \cdot \vec{v}_{BA}}{|\vec{v}_{BA}|^2}$; where $\vec{r}_{BA} = (80\hat{i} + 15\hat{j})$km

$\vec{v}_{BA} = -10\hat{i} - (30\hat{i} + 50\hat{j}) = (-40\hat{i} - 50\hat{j})$ =km/hr

$\therefore \quad t = -\frac{(80\hat{i} + 150\hat{j}) \cdot (-40\hat{i} - 50\hat{j})}{|-40\hat{i} - 50\hat{j}|^2}$

$= \frac{3200 + 7500}{4100} hr = \frac{10700}{4100}$ hr $= 2.6$hrs

13. **(2)**

$B_A = \frac{(\mu_0)(2)}{(2)(2)} \frac{3(\pi/2)}{2\pi}$ ….. (i)

$B_B = \frac{(\mu_0)(3)}{(2)(4)} \frac{5\pi/3}{2\pi}$ ….. (ii)

From (i) and (ii)

$\frac{B_A}{B_B} = \frac{6}{5}$

14. **(4)**

$\therefore \mu_1 = \frac{\rho\pi r^2 L}{L} = \mu$ and $\mu_2 = \frac{\rho 4\pi r^2 L}{L} = 4\mu$

Tension in both are the same = T

Let the speed of the wave in the wires be V_1 and V_2

$V_1 = \sqrt{\frac{T}{\mu}} = V, V_2 = \sqrt{\frac{T}{4\mu}} = \frac{V}{2}$

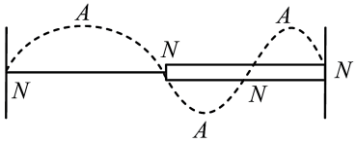
So fundamental frequencies in both wires are

$f_1 = \frac{V_1}{2L} = \frac{V}{2L}$ and $f_2 = \frac{V_2}{2L} = \frac{V}{4L}$

Frequency at which both resonate is L.C.M. of both

frequencies i.e., $\frac{V}{2L}$.

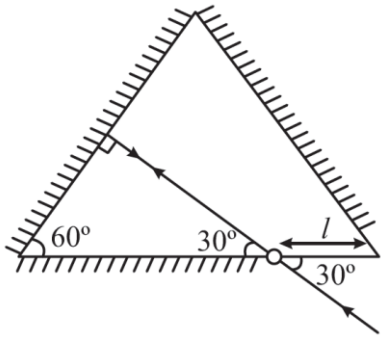
Hence no. of loops in wires are 1 and 2 respectively.



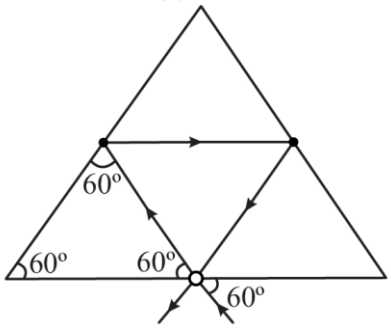
So, ratio of no. of antinodes is 1 : 2.

15. (2)

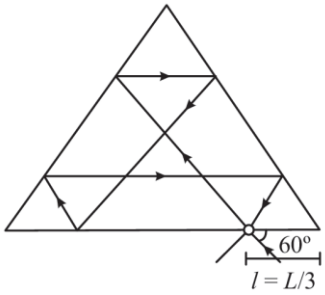
As we can see, for θ = 30° the ray will incident normally and hence will retrace its path ⇒ (*a*) is incorrect.



for θ = 60°, $l < L/2$. Then, we get ray diagram shown below. Clearly ray of light comes after two reflections ⇒ (*b*) is correct.

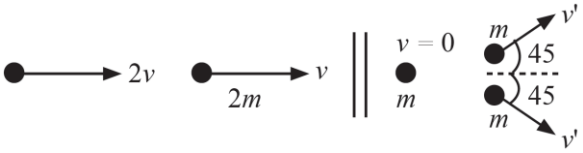


If θ = 60° and l = L/3, then we get ray diagram as shown below.



Clearly, after 5 reflections, ray comes out. So (3) and (4) are incorrect.

16. (2)



$$m2v + 2mv = m\times 0 + m\frac{v'}{\sqrt{2}}\times 2$$

$$4mv = \frac{2mv'}{\sqrt{2}} \Rightarrow v' = 2\sqrt{2}v$$

17. (3)

Let ϕ_0 be the work function of the surface of the material. Then,

According to Einstein's photoelectric equation, the maximum kinetic energy of the emitted photoelectrons in the first case is $K_{\max_1} = \frac{hc}{\lambda} - \phi_0$

and that in the second is

$$K_{\max_2} = \frac{hc}{\frac{\lambda}{2}} - \phi_0 = \frac{2hc}{\lambda} - \phi_0$$

But $K_{\max_2} = 3K_{\max_1}$ (Given)

$$\therefore \frac{2hc}{\lambda} - \phi_0 = 3\left(\frac{hc}{\lambda} - \phi_0\right)$$

$$\frac{2hc}{\lambda} - \phi_0 = \frac{3hc}{\lambda} - 3\phi_0$$

$$\therefore 3\phi_0 - \phi_0 = \frac{3hc}{\lambda} - \frac{2hc}{\lambda}$$

$$2\phi_0 = \frac{hc}{\lambda} \text{ or } \phi_0 = \frac{hc}{2\lambda}$$

18. (1)

$\Delta Q = \Delta U + \Delta W$

Heat energy exchange,

$$\Delta Q = \int_{T_0}^{\eta T_0} nc\, dT = \int_{T_0}^{\eta T_0} n\frac{\alpha}{T}\, dT$$

$= n\,\alpha \ln\eta$

Change in internal energy, $\Delta U = nC_v\Delta T$

$$= n\frac{R}{r-1}(\eta T_0 - T_0)$$

$$= \frac{nRT_0}{r-1}(\eta - 1)$$

From first law of thermodynamics

$$W = \Delta Q - \Delta U = n\,\alpha \ln\eta - \frac{nRT_0}{r-1}(\eta - 1)$$

For one mole, $n = 1$

$$W = \alpha \ln\eta - \frac{RT_0}{r-1}(\eta - 1)$$

19. (2)

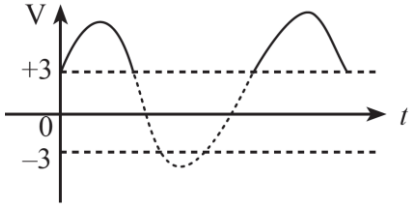
$Q = [(7.835 \times 231) + (7.07 \times 4) - (7.8 \times 235)]$

$= 5.165$ MeV

20. (2)

When $V_i > 3$ volt, $V_R > 0$, because diode will be in forward biased state.

When $V_i \leq 3$ volt, $V_R = 0$, because diode will be in reverse biased state.

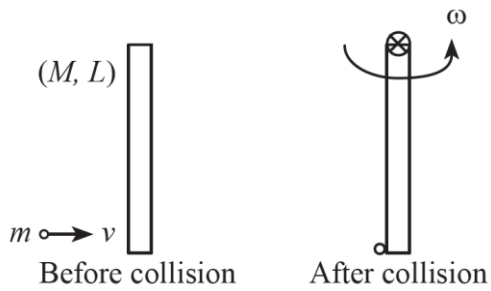


21. (20)

Here, we have given that mass of rod

$M = 0.9$kg, $\ell = 1$ m, mass of particle = 0.1 kg

Now, angular momentum is conserved because net torque is zero. $\vec{L}_i = \vec{L}_i$



$mvL = I\omega$

$mvL = \left(\frac{ML^2}{3} + mL^2\right)\omega$

$0.1'80'1 = \frac{20 \cdot 9'1^2}{c^-} + 0.1'1^2 \ddot{\theta} \varnothing WP8 = \frac{4}{10} w$

$w = 20\frac{rad}{sec}$

22. (3)

$\frac{V_{bf}}{\mu_w} = \frac{-8}{\mu_w} + \frac{(-v)}{1}$

$\Rightarrow \frac{-12}{\frac{4}{3}} = \frac{-8}{\frac{4}{3}} + \frac{(-v)}{1}$

$\Rightarrow v = 3 m/s$

23. (20)

In series connection, $R_{eq} = nR = 10n$

$\Rightarrow I_{se} = \frac{20}{10+10n} = \frac{2}{1+n}$

In parallel connection, $R_{eq} = \frac{10}{n}$

$\Rightarrow I_p = \frac{20}{\frac{10}{n}+10} = \frac{2n}{1+n}$

According to the question,

$\Rightarrow \frac{I_p}{I_{se}} = 20 = \frac{2n/1+n}{2/1+n}$

$\Rightarrow n = 20$

24. (600)

$\beta = \frac{\lambda D}{d}$

$\Delta\beta = \frac{\lambda}{d}\Delta D$

$\lambda = \frac{\Delta\beta . d}{\Delta D}$

$= \frac{3\times10^{-5}\times1\times10^{-3}}{5\times10^{-2}}$

$= 60\times10^{-8} = 600\times10^{-9} m$

$= 600 nm$

25. (220)

Least count $= \frac{\text{Pitch}}{\text{No. of circular divisions}}$

$= \frac{0.5 mm}{100}$

Least count = 5×10^{-3} mm

Positive Error = MSR + CSR (LC)

= 0 mm + $6(5 \times 10^{-3}$ mm)

Reading of Diameter = *MSR* + *CSR* (*LC*)– Positive zero error

= 4×0.5 mm + $(46(5 \times 10^{-3}))$ – $6 (5 \times 10^{-3})$ mm

= 2 mm + $40 \times 5 \times 10^{-3}$ mm = 2.2 mm

SECTION-II (CHEMISTRY)

26. (2)

In CN^-,

Total number of electrons = 14

Bond order = 3

It is diamagnetic in nature.

27. (4)

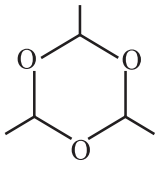
Balanced reaction involving lowest integral stoichiometric ratio

$6Fe^{2+} + Cr_2O_7^{2-} + 14H^+ \rightarrow 6Fe^{3+} + 2Cr^{3+} + 7H_2O$

a = x, y = 2b and z = 7b

c = 2z

28. (2)



29. (3)

(1) $NH_4^+ + OH^- \longrightarrow NH_3 \uparrow + H_2O$.

(2) The statement is correct ; non-chiral because there is mirror plane through the metal bisecting the dien ligand.

(3) $4Cl^-$ (s) + $Cr_2O_7^{2-}$ (s) + $6H^+$ (conc.) $\longrightarrow$ $2CrO_2Cl_2 \uparrow$ (deep red) + $3H_2O$ (l)

(4) VCl_3; $_{23}V^{3+}$;$[Ar]^{18}3d^2$ = 2 unpaired electrons.

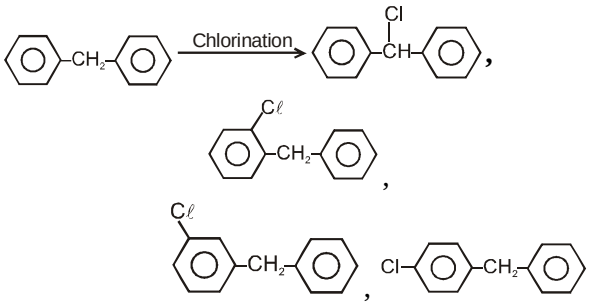
VO^{2+} ; V^{4+} ; $[Ar]^{18}3d^1$ = 1 unpaired electron.

$[VO_4]^{3-}$; V^{5+} ; $[Ar]^{18}3d^0$

= no unpaired electron.

$[V(H_2O)_6]^{2+}$; V^{2+} : $[Ar]^{18}3d^3$ = 3 unpaired electrons.

30. (1)



Total 4-isomers

31. (2)

Statement-1: HI > HBr due to low bond dissociation enthalpy.

Statement-2: Due to small size.

Statement-3: F_2 > O_3 oxidising agent. (Highest SRP of fluorine).

Statement-4: Fluorine does not have empty d-orbitals.

32. (2)

Enol (Q) is stabilised by resonance and hydrogen bond.



33. (3)

$$\underset{(X)}{2MnO_2} + 2K_2CO_3 + \underset{(air)}{O_2} \xrightarrow{\Delta} \underset{\text{green}\ (y)}{2K_2MnO_4} + 2CO_2(g)$$

$$\underset{(Y)}{2K_2MnO_2} + Cl_2 \longrightarrow \underset{(z)\ \text{Pink}}{2KMnO_4} + 2KCl$$

34. (2)

Step-III is the RDS

Overall ΔH would be negative as aromatisation takes place.

35. (3)

For Zn^{2+} ion, group reagent is H_2S in presence of NH_4OH

36. (2)

Spin quantum no. is not obtained from Schrodinger wave equation.

For uni-electronic species: energy varies with n-only.

37. (1)

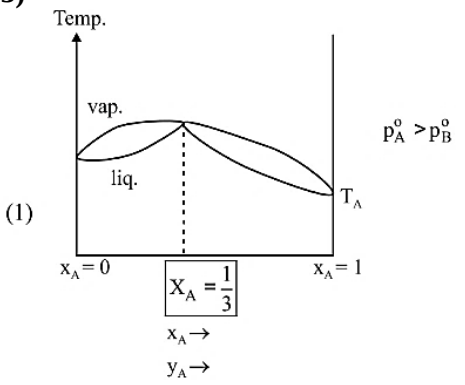
$-q_{reaction} = q_{bormb} + q_{water}$

$q_{reaction} = (C(bomb) + (m_{water} \times c))\Delta T$

$= (652 + 500 \times 4.18) \times 14.3$

$= 39210$ J or 39.21 kJ

38. (3)



(1) $x_A = \dfrac{n_A}{n_A + n_B} = \dfrac{\frac{50}{100}}{\frac{50}{100} + \frac{50}{50}} = \dfrac{1}{3}$

$x_B = \dfrac{2}{3}$ $\therefore$ A is correct statement

(3) This mix. would lie on left of azeotropic solution; hence upon distillation, the vapour would shift towards pure B.

(4) This mix. would lie in right of azeotrope & hence upon distillation, the vapour would shift towards pure A.

39. (3)

$$CH_3 - CH_2 - CH_2 - CH_2 - CH_2 - NH_2$$
$$\downarrow CH_3I / AgOH$$
$$CH_3 - CH_2 - CH_2 - CH = CH_2$$
$$\downarrow O_3 / Zn / H_2O$$
$$CH_3 - CH_2 - CH_2 - CHO + HCHO$$

40. (1)

Removal of e^- is easy from anion.

Successive I.E. is always in increasing order.

41. (2)

A→S; B→P; C→R; D→Q

42. (3)

$$CH_3CH_2CH_2C \equiv CH \xrightarrow[H^+]{H_2O} \underset{(A)}{CH_3CH_2CH_2-\overset{O}{\overset{\|}{C}}-CH_3}$$

$$\xrightarrow{NaBH_4} \underset{(B)}{CH_3CH_2CH_2-\overset{OH}{\overset{|}{CH}}-CH_3}$$

$$\xrightarrow{SOCl_2} CH_3CH_2CH_2-\underset{Cl}{\underset{|}{CH}}-CH_3 \xrightarrow{KCN\ (aq)}$$

(C)

$$CH_3CH_2CH_2-\underset{CN}{\underset{|}{CH}}-CH_3 \xrightarrow[H^+]{H_2O}$$

(D)

$$CH_3CH_2CH_2-\underset{COOH}{\underset{|}{CH}}-CH_3 \xrightarrow[Red\ P]{Br_2} CH_3CH_2CH_2-\underset{Br}{\underset{|}{\underset{COOH}{\underset{|}{C}}}}-CH_3$$

(E) (F)

43. (3)

Reaction at cathode :-

$$Cr_2O_7^{2-} + 14H^+ + 6e^- \longrightarrow 2Cr^{3+} + 7H_2O$$

1 M	–
–1/ 2 M	1 M
$\frac{1}{2}$M	1 M

Nernst equation :-

$$E_{cell} = E^o_{cell} - \frac{0.06}{n}\log_{10} Q$$

$$= 1.33 - \frac{0.06}{6}\log\frac{\left[Cr^{3+}\right]^2}{\left[Cr_2O_7^{2-}\right]\left[H^+\right]^{14}}$$

$$= 1.33 - 0.01\ \log\left[\frac{1^2}{0.5\times(0.1)^{14}}\right]$$

= 1.33 – (0.01) (14.3)

= 1.33 – 0.143

= 1.187

44. (1)

$$PhCOOH \xrightarrow{NH_3} \underset{[A]}{PhCOONH_4} \xrightarrow{Heat}$$

$$\underset{[B]}{PhCONH_2} \xrightarrow{P_2O_5} \underset{[C]}{PhCN} \xrightarrow[Ni/\Delta]{H_2} \underset{[D]}{PhCH_2NH_2}$$

$$PhCONH_2 \xrightarrow[KOH]{Br_2} PhNH_2$$

45. (1)

Covalent character $\propto$ Polarising power of cation

$\propto$ charge on cation

$Fe^{3+} > Fe^{2+}$ (Polarising power)

46. (56)

$$M_2O_3 + 3H_2 \rightarrow 2M + 3H_2O$$

3 mol of H_2 require = 1 mol of M_2O_3

$$\frac{1.2}{2}\text{ mol of } H_2 \text{ require} = \frac{1}{3}\times\frac{1.2}{2} = 0.2\text{ mol}$$

Mass of 0.2 mol = 32 g

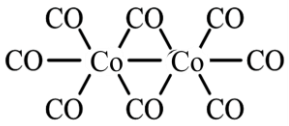
Mass of 1 mol = 160 g

2M + 48 = 160

M = 56 g

47. (7)

Structure of $[Co_2(CO)_8]$



Number of Co – Co bonds = 1 = X

Number of terminal CO ligands = 6 = Y

$\therefore$ X + Y = 1 + 6 = 7

48. (5)

Only gly is optically inactive

x = 5 ; y = 1

49. (14)

	TCl	+ NaOT	⟶	NaCl + T_2O
Initial millimoles	2	3.75		

Millimoles of NaOT left = 1.75

$$[OT^-] = \frac{1.75}{25} = 7\times10^{-2}\text{ M}$$

$$pOT = 2 - \log 7$$

pT + pOT = (7.62) × 2

pT = 15.24 – pOT

= 15.24 – 2 + log 7

= 13.24 + log 7

= 13.24 + 0.76 = 14.085

50. (526)

$$k = Ae^{-E_a/RT}$$

$$\log k = \log A - \frac{E_a}{2.303R}\times\frac{1}{T}$$

From given graph : slope = – 10000 = $\frac{E_a}{2.303R}$

$$\because \log\frac{k_2}{k_1} = \frac{E_a}{2.303\,R}\left[\frac{1}{T_1} - \frac{1}{T_2}\right]$$

$$\log\frac{10^{-4}}{10^{-5}} = 10000\left[\frac{1}{500} - \frac{1}{T_2}\right]$$

$$\Rightarrow 1 = 20 - \frac{10000}{T_2}$$

$$\Rightarrow T_2 = \frac{10000}{19} = 526.3 \approx 526\text{ K}$$

SECTION-III (MATHEMATICS)

51. (1)

(A is an idempotent matrix)

$\because \quad A^2 = A$

$(I + A)^4 = (I + A)^2(I + A)^2$

$= (I^2 + A^2 + 2IA)(I^2 + A^2 + 2IA)$

$= (I + A + 2A)(I + A + 2A)$

$= (3A + I)(3A + I)$

$= 9A^2 + 3AI + 3AI + I^2 = 15A + I$

52. (4)

$$\frac{1}{2^6}\left[{}^7C_1 + {}^7C_3(4x+1) + {}^7C_5(4x+1)^2 + {}^7C_7(4x+1)^3\right]$$

53. (3)

$4x^2 - 15x + \lambda = 0$

$f(1) > 0$ and $f(2) < 0$ and $f(3) > 0$

$f(1) = 4 - 15 + \lambda > 0$

$\lambda > 11$

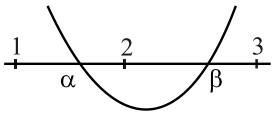
$f(2) = 16 - 30 + \lambda < 0$

$\lambda < 14$

$f(3) = 36 - 45 + \lambda > 0$

$\lambda > 9$

So, $\lambda = 12, 13$ are 2 integral solutions.



54. (2)

From given differential equation

$$\frac{dy}{dx} = \frac{(x-2)^2 + (y-2)^2}{(x-2)(y-2)} \qquad \ldots(i)$$

Let $x - 2 = X$ & $y - 2 = Y \Rightarrow \frac{dY}{dX} = \frac{dy}{dx}$

$\therefore$ From equation (i)

$$\frac{dY}{dX} = \frac{X^2 + Y^2}{XY} \qquad \ldots(ii)$$

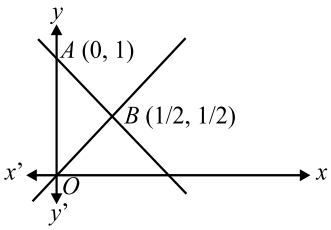
(Homogeneous differential equation)

Put $Y = VX$, then $\frac{dY}{dX} = V + X\frac{dV}{dX}$

Substituting these values in (ii) and then solving we get, $\frac{Y^2}{X^2} = 2\ln|X| + c$

$$\Rightarrow \quad \frac{(y-2)^2}{(x-2)^2} = 2\ln|x-2| + c$$

55. (1)



The given lines are $x(x + y - 1)(x - y) = 0$. In other words, lines $x = 0$, $x + y - 1 = 0$ and $x - y = 0$ form ΔOAB as shown in the above diagram

The triangle is right angled at point B, hence orthocentre is (1/2, 1/2). Also, circumcentre is mid-point of OA which is (0, 1/2). The centroid is

$$\left(\frac{0+\frac{1}{2}+0}{3}, \frac{0+\frac{1}{2}+1}{3}\right) = \left(\frac{1}{6}, \frac{1}{2}\right)$$

Also, $OA = 1$, $OB = OC = \frac{1}{\sqrt{2}}$. Hence, the incentre is

$$\left(\frac{0\left(\frac{1}{\sqrt{2}}\right) + \frac{1}{2}(1) + 0\left(\frac{1}{\sqrt{2}}\right)}{\frac{1}{\sqrt{2}} + 1 + \frac{1}{\sqrt{2}}}, \frac{0\left(\frac{1}{\sqrt{2}}\right) + \frac{1}{2}(1) + 1\left(\frac{1}{\sqrt{2}}\right)}{\frac{1}{\sqrt{2}} + 1 + \frac{1}{\sqrt{2}}}\right)$$

$$= \left(\frac{1}{2+2\sqrt{2}}, \frac{1}{2}\right)$$

56. (1)

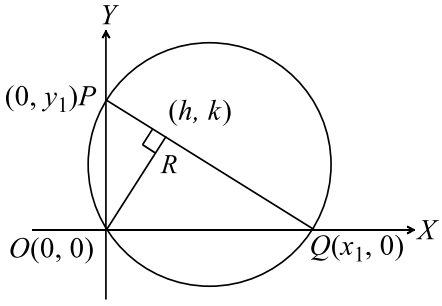
$$S.D. = \left|\frac{(\vec{b} - \vec{a}) \cdot (\vec{p} \times \vec{q})}{|\vec{p} \times \vec{q}|}\right|$$

$$S.D. = \left|\frac{\left(-3\hat{j} - \hat{k}\right) \cdot \left(\hat{i} + 2\hat{j} - 3\hat{k}\right)}{\sqrt{14}}\right|$$

$$d = \left|\frac{-6+3}{\sqrt{14}}\right| = \frac{3}{\sqrt{14}}$$

$\sqrt{14}d = 3 \Rightarrow 14d^2 = 9$

57. (3)



Equation of line PQ is $y-k=-\frac{h}{k}(x-h)$

or $hx+ky=h^2+k^2$

$\Rightarrow$ Points $Q\left(\frac{h^2+k^2}{h},0\right)$ and $P\left(0,\frac{h^2+k^2}{k}\right)$

Also $2a=\sqrt{x_1^2+y_1^2}$

$\Rightarrow x_1^2+y_1^2=4a^2$

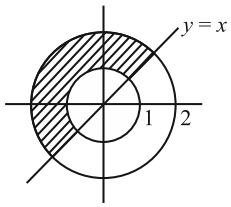
Eliminating x_1 and y_1 we have

$\left(x^2+y^2\right)^2\left(\frac{1}{x^2}+\frac{1}{y^2}\right)=4a^2$

58. (3)

$k=\frac{\pi(2^2-1^2)}{2}=\frac{3\pi}{2}$

$\int_0^{\pi}\{\sin x\}dx=\int_0^{\pi}\sin x dx=2$



59. (3)

$^{(100-55)+(10-1)}C_{10-1}={}^{54}C_9$

60. (4)

Sum $=\sum_{n=1}^{n}\tan^{-1}\left[\frac{\sqrt{2n+2}-\sqrt{2n-2}}{2+\sqrt{n^2-1}}\right]$

$=\sum_{n=1}^{n}\tan^{-1}\left[\frac{\sqrt{\frac{n+1}{2}}-\sqrt{\frac{n-1}{2}}}{1+\sqrt{\frac{n+1}{2}}\cdot\sqrt{\frac{n-1}{2}}}\right]$ (divided by 2)

$=\sum_{n=1}^{n}\left(\tan^{-1}\sqrt{\frac{n+1}{2}}-\tan^{-1}\sqrt{\frac{n-1}{2}}\right)$

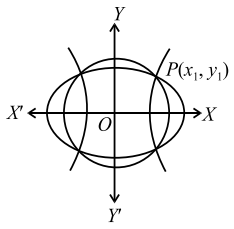
$=\tan^{-1}\left(\sqrt{\frac{n+1}{2}}\right)+\tan^{-1}\sqrt{\frac{n}{2}}-\tan^{-1}\frac{1}{\sqrt{2}}$

$\lim_{n\to\infty}\text{sum}=\pi-\tan^{-1}\left(\frac{1}{\sqrt{2}}\right)=\frac{\pi}{2}+\cot^{-1}\left(\frac{1}{\sqrt{2}}\right)$

$=\frac{\pi}{2}+\tan^{-1}(\sqrt{2})$

$\therefore\ k=\sqrt{2}\Rightarrow k^2=2$

61. (1)



Since ellipse and hyperbola intersect orthogonally, they are confocal

Hence, $a = 2$ (equating foci)

Let point of intersection in the first quadrant be $P(x_1, y_1)$. P lies on both the curves. Therefore,

$4x_1^2 + 9y_1^2 = 36$ and $4x_1^2 - y_1^2 = 4$

Adding these two results, we get

$8(x_1^2 + y_1^2) = 40$

$\Rightarrow x_1^2 + y_1^2 = 5 \Rightarrow r = \sqrt{5}$

Hence, equation of the circle is

$x^2 + y^2 = 5$

62. (1)

$f'(x)=\sec x-\text{cosec}\,x$

$f'(x)=0$ at $x=\frac{\pi}{4}$

Also $f''\left(\frac{\pi}{4}\right)>0$ so $f(x)$ is minimum at $x=\frac{\pi}{4}$

$f\left(\frac{\pi}{4}\right)=\int_0^{\pi/4}\sec\theta\,d\theta+\int_{\pi/4}^{\pi/2}\text{cosec}\,\theta\,d\theta=\ln\left(\frac{\sqrt{2}+1}{\sqrt{2}-1}\right)$

$=2\ln(\sqrt{2}+1)$

63. (1)

$\vec{a}=\lambda\hat{i}+2\hat{j}-3\hat{k}$

$\vec{b}=\hat{i}-\lambda\hat{j}+2\hat{k}$

$\Rightarrow\left(\left(\vec{a}-\vec{b}\right).\left(\vec{a}+\vec{b}\right)\right)\left(\vec{a}\times\vec{b}\right)=8\hat{i}-40\hat{j}-24\hat{k}$

$\Rightarrow 8\left(\vec{a}\times\vec{b}\right)=8\hat{i}-40\hat{j}-24\hat{k}$

Now, $\vec{a}\times\vec{b}=\begin{vmatrix}\hat{i}&\hat{j}&\hat{k}\\ \lambda&2&-3\\ 1&-\lambda&2\end{vmatrix}$

$=(4-3\lambda)\hat{i}-(2\lambda+3)\hat{j}+(-\lambda^2-2)\hat{k}$

$\Rightarrow\lambda=1$

$\therefore\ \vec{a}=\hat{i}+2\hat{j}-3\hat{k}$

$\vec{b}=\hat{i}-\hat{j}+2\hat{k}$

$\Rightarrow\vec{a}+\vec{b}=2\hat{i}+\hat{j}-\hat{k},\vec{a}-\vec{b}=3\hat{j}-5\hat{k}$

$\Rightarrow\left(\vec{a}+\vec{b}\right)\times\left(\vec{a}-\vec{b}\right)\begin{vmatrix}\hat{i}&\hat{j}&\hat{k}\\ 2&1&-1\\ 0&3&-5\end{vmatrix}=-2\hat{i}+10\hat{j}+6\hat{k}$

$\therefore$ Required answer = 4 + 100 + 36 = 140

64. (1)

$I=\int_{\alpha}^{\beta}\frac{e^{f(a(\alpha-x))}}{e^{f(a(\alpha-x))}+e^{f(a(\beta-x))}}dx \qquad ...(1)$

$I=\int_{\alpha}^{\beta}\frac{e^{f(a(x-\beta))}}{e^{f(a(\alpha-x))}+e^{f(a(\beta-x))}}dx \qquad ...(2)$

From (1) and (2), we get

$\Rightarrow I=\frac{1}{2}(\beta-\alpha)$ ($\because f(-x)=f(x)$)

65. (3)

pi must be divisible by 2 and 3 both. Hence $p_1 = 6$, $p_2 = 2$, $p_3 = 3$.

66. (4)

Focus is $S \equiv (2, 0)$. Points $P \equiv (0, 0)$ and $Q = (2t^2, 4t)$

$$\text{Area of } PQS = \frac{1}{2}\begin{vmatrix} 0 & 0 & 1 \\ 2 & 0 & 1 \\ 2t^2 & 4t & 1 \end{vmatrix} = \frac{1}{2}(8t) = 4t \ldots.(i)$$

$Q\,(2t^2, 4t)$ satisfies circle

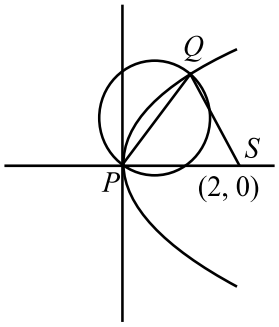
$4t^4 + 16t^2 - 4t^2 - 16t = 0$

$t^3 + 3t - 4 = 0$

$(t - 1)(t^2 + t + 4) = 0$

Put $t = 1$ in area of PQS

$\Rightarrow$ Area of PQS is 4



67. (2)

$$\int \tan x \tan 2x \tan 3x\, dx$$

$$= \int (\tan 3x - \tan x - \tan 2x)\, dx$$

$$\left(\because \tan 3x = \frac{\tan 2x + \tan x}{1 - \tan 2x \cdot \tan x}\right)$$

$$= -\frac{1}{3}\ell n|\cos 3x| + \ell n|\cos x| + \frac{1}{2}\ell n|\cos 2x| + C$$

$$= -\ell n|\sec x| + \frac{1}{3}\ell n|\sec 3x| - \frac{1}{2}\ell n|\sec 2x| + C$$

68. (2)

$$\sigma^2 = \left(\frac{\sum x_i^2}{n} - \left(\frac{\sum x_i}{n}\right)^2\right) = \left(\frac{\sum n^2}{n} - \left(\frac{\sum x}{n}\right)^2\right)$$

$$= \frac{n^2 - 1}{12}$$

So, standard deviation = $\sqrt{\frac{n^2 - 1}{12}}$

Mean $(\bar{x}) = \frac{\sum x_i}{n} = \frac{n(n+1)}{2n} = \frac{(n+1)}{2}$

So coefficient of variation

$$= \frac{\sigma}{\bar{x}} \times 100 = \sqrt{\frac{(n-1)}{3(n+1)}} \times 100$$

69. (1)

Since $0 < \theta < \frac{\pi}{4}$, $0 < \sin^2\theta, \cos^2\theta, \tan^2\theta < 1$

(A) $\sum_{k=0}^{\infty} \sin^{2k}\theta = \frac{1}{1 - \sin^2\theta} = \sec^2\theta$

(B) $\sum_{k=0}^{\infty} \cos^{2k}\theta = \frac{1}{1 - \cos^2\theta} = \text{cosec}^2\theta$

(C) $\sum_{k=0}^{\infty} \tan^{2k}\theta = \frac{1}{1 - \tan^2\theta} = \frac{\cos^2\theta}{\cos^2\theta - \sin^2\theta}$

$$= \frac{\cos^2\theta}{\cos 2\theta}$$

(D) $\cot\theta > 1$ in $0 < \theta < \frac{\pi}{4}$.

Hence $\sum_{k=0}^{\infty} \cot^{2k}\theta$ does not exist

70. (4)

$$\sum_{i=1}^{\infty} a_i = \frac{2\cot x}{1 - \sin^2 x} = \frac{2}{\cos x \sin x}$$

$$\sum_{j-1}^{\infty} b_i = \frac{\sin 2x}{1 - \sin^2 x} = \frac{2\sin x}{\cos x}$$

$$\therefore \quad \sum_{i=1}^{\infty} a_i - \sum_{j=1}^{\infty} b_i = \frac{2}{\cos x \cdot \sin x} = \frac{2\sin x}{\cos x}$$

$$= \frac{2\left(1 - \sin^2 x\right)}{\cos x \sin x} = 2\cot x$$

$\therefore$ Minimum value = 2

71. (15)

$z = 1 + i$

$$z_1 = \frac{1 + i\bar{z}}{\bar{z}(1 - z) + \frac{1}{z}}$$

$$z_1 = \frac{1 + i(1 - i)}{(1 - i)(1 - 1 - i) + \frac{1}{1 + i}}$$

$$= \frac{1 + i - i^2}{(1 - i)(-i) + \frac{1 - i}{2}}$$

$$= \frac{2 + i}{\frac{-3i - 1}{2}} = \frac{4 + 2i}{-3i - 1}$$

$$= \frac{-(4 + 2i)(3i - 1)}{(3i)^2 - (1)^2}$$

Arg $(z_1) = \frac{3\pi}{4}$

$$\therefore \frac{20}{\pi}\arg(z_1) = \frac{20}{\pi} \times \frac{3\pi}{4} = 15$$

72. **(31)**

$$\frac{(x-3)^{\frac{-|x|}{x}}\sqrt{(x-4)^2}\,(17-x)}{\sqrt{-x}\left(-x^2+x-1\right)\left(|x|-32\right)}<0$$

For $\sqrt{-x}$ to be defined $x < 0$

$$\Rightarrow \frac{-|x|}{x}=1$$

Since $\sqrt{(x-4)^2}$, $\sqrt{-x}$ are positive & $-x^2+x-1$ is always negative

$$\frac{(x-3)(x-17)}{x+32}>0$$

$x < 0$

$$\Rightarrow \frac{1}{x+32}>0$$

$x > -32$

$x \in (-32, 0)$

73. **(4)**

Let

$E_1 \rightarrow$ the event when die A is used

$E_2 \rightarrow$ the event when die B is used

$R \rightarrow$ when red face appears

$$P\left(\frac{R}{E_1}\right)=\left(\frac{3}{6}\right)^n \text{ and } P\left(\frac{R}{E_2}\right)=\left(\frac{2}{6}\right)^n$$

$$P(E_1)=\frac{1}{2},\ P(E_2)=\frac{1}{2}$$

$$\therefore \quad P\left(\frac{E_2}{R}\right)=\frac{\frac{1}{2}\cdot\left(\frac{2}{6}\right)^n}{\left[\left(\frac{3}{6}\right)^n+\left(\frac{2}{6}\right)^n\right]\cdot\frac{1}{2}}=\frac{16}{97}$$

$$\Rightarrow \quad \frac{2^n}{3^n+2^n}=\frac{16}{97} \quad \Rightarrow \quad n=4$$

74. **(1)**

$$A=\lim_{x\to 0}\frac{\left(x-\frac{x^3}{\lfloor 3}+....\right)+a\left(1+x+\frac{x^2}{\lfloor 2}+\frac{x^3}{\lfloor 3}+...\right)+b\left(1-x+\frac{x^2}{\lfloor 2}-\frac{x^3}{\lfloor 3}+...\right)+c\left(x-\frac{x^2}{2}+\frac{x^3}{3}.....\right)}{x^3}$$

$$=\frac{(a+b)+(1+a-b+c)x+\left(\frac{a}{2}+\frac{b}{2}-\frac{c}{2}\right)x^2+\left(-\frac{1}{\lfloor 3}+\frac{a}{\lfloor 3}-\frac{b}{\lfloor 3}+\frac{c}{\lfloor 3}\right)x^3+...}{x^3}$$

So far the limit to be finite

$$\left.\begin{aligned} a+b&=0\\ 1+a-b+c&=0\\ \frac{a}{2}+\frac{b}{2}-\frac{c}{2}&=0\end{aligned}\right\}\Rightarrow a=-1/2,\ b=1/2 \text{ and } c=0$$

and $\frac{-1}{\lfloor 3}+\frac{a}{\lfloor 3}-\frac{b}{\lfloor 3}+\frac{c}{3}=A$

$$\Rightarrow A=-\frac{1}{6}-\frac{1}{12}-\frac{1}{12}+0=\frac{-1}{3}\text{(finite value)}$$

Now, $[|y + a| - b] \le 2c$

$\Rightarrow [|y-1/2|-1/2] \le 0$

$\Rightarrow |y-1/2|-1/2<1$

$\Rightarrow |y-1/2|<3/2$

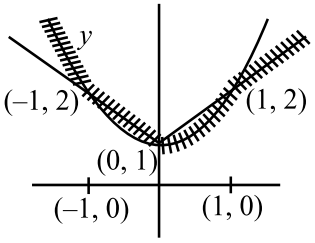
$\Rightarrow -3/2<y-1/2<3/2$

$\Rightarrow -1<y<2$

75. **(3)**

$$f(x)=|x|+1=\begin{cases}x+1 & x\ge 0\\ -x+1 & x<0\end{cases}$$

$g(x) = x^2 + 1$



Number of non-differentiable points = 3.



PW Web/App - https://smart.link/7wwosivoicgd4

Library- https://smart.link/sdfez8ejd80if